Table 4.1. The ranges and average values of the selected physicochemical properties of the soil samples obtained from the wetlands, grasslands and croplands land use types across the Lowlands, Foothills, Mountains and Senqu river valley zones of Lesotho, *together with the coefficient of variation (% CV) given in parentheses*.

Soil property	Minimum	Maximum	Average (CV)	Variability class
		Physical properties		
Sand (%)	7.28	79.95	56.80 (23)	Moderately low
Silt (%)	6.67	61.33	22.30 (35)	Moderate
Clay (%)	9.39	74.05	20.89 (50)	Moderately high
OC (%)	0.10	25.39	4.13 (106)	Extremely high
Bulk density (kg/m³)	1011.63	2038.72	1588.10 (14)	Low
		Chemical properties		
pH_{water}	5.04	7.57	6.07 (8)	Low
pH_{KCl}	4.13	7.03	5.21 (10)	Low
EC (dS/m)	0.02	0.23	0.07 (56)	High
CEC (cmol/kg)	3.13	141.12	42.62 (68)	High
Base saturation (%)	1.71	141.21	24.58 (87)	Very high
ESP (%)	0.01	1.63	0.21 (139)	Extremely high
SAR	0.00	0.37	0.05 (117)	Extremely high
		Macronutrients		
Total N (%)	0.04	1.03	0.21 (94)	Very high
Available P (mg/kg)	0.01	1.33	0.20 (108)	Extremely high
Exchangeable Ca^2 (mg/kg)	30.88	4941.67	1285.04 (86)	Very high
Exchangeable Mg^2 (mg/kg)	8.43	3760.83	405.37 (166)	Extremely high
Exchangeable K (mg/kg)	0.74	331.40	85.80 (96)	Extremely high
		Micronutrients and trace elements		
Exchangeable Na (mg/kg)	0.61	224.20	21.57 (167)	Extremely high
Extractable Fe (mg/kg)	0.36	40.62	4.03 (134)	Extremely high
Extractable Mn (mg/kg)	0.22	17.50	3.16 (124)	Extremely high
Extractable Cu (mg/kg)	0.10	9.75	3.40 (67)	High
Extractable Zn (mg/kg)	0.52	7.22	2.99 (60)	High

Variability class is based on the coefficient of variation (CV, %): Low = <20, Moderately low = 20–34, Moderate = 35–44, Moderately high = 45–54, High = 55–74, Very high = 75–94, Extremely high = ≥95.

Table 4.2. The ranges and average values of the selected physicochemical properties of the study soil samp agroecological zones and (b) land use types, *together with the coefficient of variation (% CV) given in parenth*

		Physical properties				
Agroecological zones		**Sand (%)**	**Silt (%)**	**Clay (%)**	**OC (%)**	
Lowlands (n = 17)	Minimum	7.28	6.67	12.05	0.10	
	Maximum	79.95	61.33	74.05	4.69	
	Average (CV)	51.95 (37)	21.37 (61)	26.68 (52)	1.19 (109)	
	Variability class	Moderate	High	Moderately high	Extremely high	
Foothills (n = 8)	Minimum	45.95	14.67	14.05	0.66	
	Maximum	68.61	23.33	35.39	6.57	
	Average (CV)	55.36 (15)	19.08 (18)	25.55 (28)	2.40 (89)	
	Variability class	Low	Low	Moderately low	Very high	
Mountains (n = 38)	Minimum	30.61	13.33	9.39	0.60	
	Maximum	74.61	34.00	42.72	25.39	
	Average (CV)	59.42 (17)	23.65 (24)	16.93 (46)	6.39 (74)	
	Variability class	Low	Moderately low	Moderately high	High	
Senqu River Valley (n = 7)	Minimum	35.28	18.00	12.05	0.13	
	Maximum	69.28	26.67	40.72	2.24	
	Average (CV)	56.04 (19)	20.95 (15)	23.01 (41)	0.95 (78)	
	Variability class	Low	Low	Moderate	Very high	

		Chemical properties					
Agroecological zones		**pH_{water}**	**pH_{KCl}**	**EC (dS/m)**	**CEC (cmol/kg)**	**Base saturation (%)**	**E (**
Lowlands (n = 17)	Minimum	5.28	4.27	0.02	3.13	4.66	0
	Maximum	7.57	7.03	0.23	56.64	141.21	1
	Average (CV)	6.12 (9)	5.20 (14)	0.06 (82)	27.97 (53)	27.34 (117)	0.40
	Variability class	Low	Low	Very high	Moderately high	Extremely high	Very
Foothills (n = 8)	Minimum	5.04	4.13	0.04	4.94	7.66	0
	Maximum	6.65	5.82	0.12	62.47	56.48	0
	Average (CV)	6.11 (10)	5.10 (12)	0.07 (44)	32.64 (63)	25.52 (72)	0.14
	Variability class	Low	Low	Moderate	High	High	Extr h
Mountains (n = 38)	Minimum	5.36	4.36	0.03	17.05	1.71	0
	Maximum	6.84	5.64	0.16	141.12	74.41	1
	Average (CV)	5.96 (6)	5.12 (6)	0.08 (46)	54.56 (59)	24.77 (70)	0.17

								0.05 (140)
	Variability class	Low	Low	Moderately high	High	High	Extremely high	Extremely high
Senqu River Valley (n = 7)	Minimum	6.09	5.34	0.03	11.18	4.68	0.02	0.01
	Maximum	6.89	6.29	0.05	38.84	32.54	0.31	0.06
	Average (CV)	6.53 (4)	5.86 (5)	0.04 (15)	24.78 (42)	15.81 (71)	0.12 (83)	0.03 (66)
	Variability class	Low	Low	Low	Moderate	High	Very high	High

Macronutrients

Agroecological zones		**Total N (%)**	**Available P (mg/kg)**	**Exchangeable Ca^2 (mg/kg)**	**Exchangeable Mg^2 (mg/kg)**	**Exchangeable K (mg/kg)**
Lowlands (n = 17)	Minimum	0.04	0.01	30.88	9.94	4.53
	Maximum	0.38	0.70	2689.67	3760.83	331.40
	Average (CV)	0.10 (99)	0.25 (77)	786.21 (95)	537.78 (173)	125.65 (85)
	Variability class	Extremely high	Very high	Extremely high	Extremely high	Very high
Foothills (n = 8)	Minimum	0.04	0.02	138.67	10.17	12.36
	Maximum	0.33	0.68	3972.50	2507.08	157.89
	Average (CV)	0.16 (79)	0.21 (130)	1251.55 (108)	497.74 (185)	71.59 (72)
	Variability class	Very high	Extremely high	Extremely high	Extremely high	High
Mountains (n = 38)	Minimum	0.04	0.03	49.61	8.43	6.77
	Maximum	1.03	0.65	4941.67	1952.00	290.27
	Average (CV)	0.30 (72)	0.16 (82)	1636.77 (70)	379.64 (140)	78.22 (88)
	Variability class	High	Very high	High	Extremely high	Very high
Senqu River Valley (n = 7)	Minimum	0.04	0.04	105.08	19.84	0.74
	Maximum	0.13	1.33	1888.75	348.13	263.33
	Average (CV)	0.06 (60)	0.30 (154)	625.36 (101)	117.93 (102)	46.38 (207)
	Variability class	High	Extremely high	Extremely high	Extremely high	Extremely high

Micronutrients and trace elements

Agroecological zones		Exchangeable Na (mg/kg)	Extractable Fe (mg/kg)	Extractable Mn (mg/kg)	Extractable Cu (mg/kg)	Extractable Zn (mg/kg)
Lowlands (n = 17)	Minimum	0.67	0.51	0.42	0.10	0.62
	Maximum	127.11	12.12	14.69	8.65	6.54
	Average (CV)	31.11 (117)	2.96 (118)	3.37 (117)	2.71 (92)	2.24 (71)
	Variability class	Extremely high	Extremely high	Extremely high	Very high	High
Foothills (n = 8)	Minimum	0.61	0.58	0.67	2.22	1.46
	Maximum	59.25	40.62	17.50	9.75	6.13
	Average (CV)	13.12 (154)	7.27 (188)	3.65 (155)	5.29 (54)	3.34 (55)
	Variability class	Extremely high	Extremely high	Extremely high	Moderately high	High
Mountains (n = 38)	Minimum	1.09	0.36	0.25	0.78	0.68
	Maximum	224.20	12.42	14.28	8.46	7.22
	Average (CV)	21.80 (187)	4.16 (74)	2.97 (119)	3.42 (58)	3.51 (50)
	Variability class	Extremely high	High	Extremely high	High	Moderately high
Senqu River Valley (n = 7)	Minimum	1.77	0.43	0.22	0.64	0.52
	Maximum	23.13	5.75	13.66	5.41	3.43
	Average (CV)	6.82 (113)	2.26 (89)	3.12 (151)	2.78 (64)	1.55 (85)
	Variability class	Extremely high	Very high	Extremely high	High	Very high

Physical properties

Land use types		Sand (%)	Silt (%)	Clay (%)	OC (%)	Bulk density (kg/m³)
Croplands (n = 16)	Minimum	30.61	6.67	13.39	0.10	1468.17
	Maximum	79.95	33.33	42.72	7.41	2038.72
	Average (CV)	50.49 (25)	21.88 (31)	27.64 (32)	1.49 (118)	1791.54 (8)
	Variability class	Moderately low	Moderately low	Moderately low	Extremely high	Low
Grasslands (n = 33)	Minimum	51.28	10.67	9.39	0.23	1102.12
	Maximum	74.61	34.00	28.72	25.39	1957.96

	Average (CV)	62.45 (10)	21.29 (28)	16.26 (33)	4.55 (106)	1556.03 (14)
	Variability class	Low	Moderately low	Moderately low	Extremely high	Low
	Minimum	7.28	9.33	11.39	0.13	1011.63
	Maximum	72.61	61.33	74.05	18.13	1845.99
Wetlands (n = 21)	Average (CV)	52.74 (32)	24.22 (44)	23.04 (61)	5.47 (79)	1483.49 (14)
	Variability class	Moderately low	Moderate	High	Very high	Low

Chemical properties

Land use types		**pH_{water}**	**pH_{KCl}**	**EC (dS/m)**	**CEC (cmol/kg)**	**Base saturation (%)**	**ESP (%)**	**SAR**
	Minimum	5.04	4.13	0.02	4.94	5.16	0.02	0.01
	Maximum	6.84	5.84	0.09	141.12	53.11	0.31	0.10
Croplands (n = 16)	Average (CV)	6.16 (9)	5.14 (11)	0.04 (34)	38.55 (88)	19.93 (73)	0.14 (69)	0.04 (73)
	Variability class	Low	Low	Moderately low	Very high	High	High	High
	Minimum	5.28	4.27	0.02	3.13	1.71	0.01	0.00
	Maximum	6.71	6.05	0.15	120.50	74.41	0.40	0.12
Grasslands (n = 33)	Average (CV)	6.05 (6)	5.22 (9)	0.07 (53)	36.29 (63)	22.98 (77)	0.12 (84)	0.03 (87)
	Variability class	Low	Low	Moderately high	High	Very high	Very high	Very high
	Minimum	5.36	4.48	0.02	14.19	4.39	0.03	0.02
	Maximum	7.57	7.03	0.23	121.05	141.21	1.63	0.37
Wetlands (n = 21)	Average (CV)	6.05 (9)	5.25 (11)	0.09 (50)	55.67 (55)	30.64 (95)	0.43 (110)	0.10 (96)
	Variability class	Low	Low	Moderately high	High	Extremely high	Extremely high	Extremely high

Macronutrients

Land use types		**Total N (%)**	**Available P (mg/kg)**	**Exchangeable Ca^2 (mg/kg)**	**Exchangeable Mg^2 (mg/kg)**	**Exchangeable K (mg/kg)**
Croplands (n = 16)	Minimum	0.04	0.01	32.42	10.17	12.14
	Maximum	0.28	1.33	4941.67	1952.00	279.47
	Average (CV)	0.08 (85)	0.31 (108)	1121.36 (120)	338.87 (159)	85.17 (100)

	Variability class	Very high	Extremely high	Extremely high	Extremely high	Extremely high
Grasslands (n = 33)	Minimum	0.04	0.02	30.88	9.94	4.53
	Maximum	1.03	0.70	3260.83	2507.08	331.40
	Average (CV)	0.22 (98)	0.19 (88)	1075.78 (81)	292.05 (191)	88.72 (89)
	Variability class	Extremely high	Very high	Very high	Extremely high	Very high
Wetlands (n = 21)	Minimum	0.05	0.02	255.25	8.43	0.74
	Maximum	0.80	0.36	4815.83	3760.83	314.67
	Average (CV)	0.31 (65)	0.12 (89)	1738.58 (67)	634.12 (138)	81.70 (110)
	Variability class	High	Very high	High	Extremely high	Extremely high

Micronutrients and trace elements

Land use types		**Exchangeable Na (mg/kg)**	**Extractable Fe (mg/kg)**	**Extractable Mn (mg/kg)**	**Extractable Cu (mg/kg)**	**Extractable Zn (mg/kg)**
Croplands (n = 16)	Minimum	0.61	0.36	0.26	0.10	0.62
	Maximum	62.47	5.66	14.28	8.13	5.79
	Average (CV)	13.05 (127)	1.85 (75)	2.57 (136)	3.15 (58)	2.40 (64)
	Variability class	Extremely high	Very high	Extremely high	High	High
Grasslands (n = 33)	Minimum	0.67	0.43	0.22	0.19	0.52
	Maximum	44.41	40.62	17.50	9.75	6.25
	Average (CV)	9.37 (107)	3.91 (177)	3.73 (119)	2.67 (74)	3.01 (58)
	Variability class	Extremely high	Extremely high	Extremely high	High	High
Wetlands (n = 21)	Minimum	4.84	0.66	0.42	0.58	0.54
	Maximum	224.20	12.42	13.66	8.65	7.22
	Average (CV)	47.24 (118)	5.89 (65)	2.72 (124)	4.73 (54)	3.39 (60)
	Variability class	Extremely high	High	Extremely high	Moderately high	High

Variability class is based on the coefficient of variation (CV, %): Low = <20, Moderately low = 20–34, Moderate = 35–44, Moderately high = 45–54, High = 55–74, Very high = 75–94, Extremely high = ≥95.

Table 4.3. Summary of the Kruskal–Wallies H test results showing the significance levels for the variation in soil physicochemical characteristics as a function of AEZs and LUTs in Lesotho.

	Source of variation		
Soil property	**AEZ (df = 3)**	**LUT (df = 2)**	**AEZ × LUT (11)**
Sand (%)	0.35523^{NS}	0.00063***	0.00375**
Silt (%)	0.05358†	0.63781^{NS}	0.15166^{NS}
Clay (%)	0.00078***	0.00010****	0.00006****
OC (%)	0.00000****	0.00350**	0.00000****
Bulk density (kg/m³)	0.00038***	0.00004****	0.00002****
pH_{water}	0.01066*	0.39005^{NS}	0.00418**
pH_{KCl}	0.00437**	0.89340^{NS}	0.01752*
EC (dS/m)	0.01388*	0.00023***	0.00189**
CEC (cmol/kg)	0.00129**	0.03312*	0.00299**
Base saturation (%)	0.70763^{NS}	0.39397^{NS}	0.25154^{NS}
ESP (%)	0.00125**	0.01233*	0.00644**
SAR	0.00225**	0.00091***	0.00493**
Total N (%)	0.00003****	0.00021***	0.00003****
Available P (mg/kg)	0.49989^{NS}	0.49868^{NS}	0.65602^{NS}
Exchangeable Ca^{2}	0.00678**	0.03247*	0.00697**
Exchangeable Mg^{2}	0.28145^{NS}	0.05313†	0.16760^{NS}
Exchangeable K	0.05194†	0.87429^{NS}	0.34099^{NS}
Exchangeable Na	0.06855†	0.00059***	0.02733*
Extractable Fe	0.09946†	0.00054***	0.00532**
Extractable Mn	0.94683^{NS}	0.73372^{NS}	0.83160^{NS}

Extractable Cu	0.12056^{NS}	0.00681**	0.03950*
Extractable Zn	0.00484**	0.31421^{NS}	0.04013*

Figures

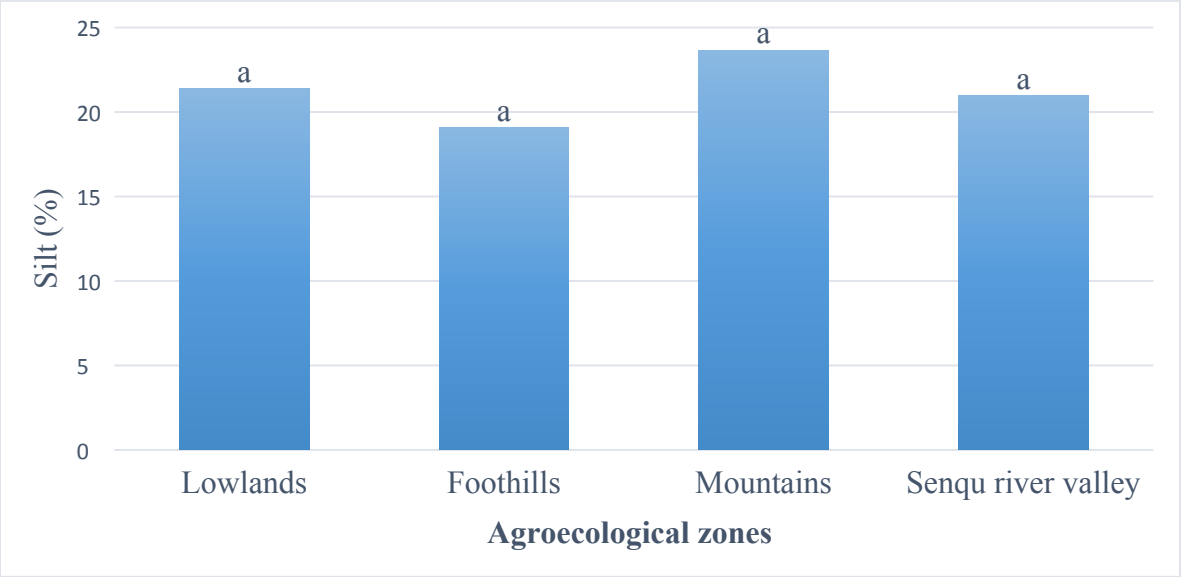


Figure 4.1. Average silt fraction across agroecological zones of Lesotho.

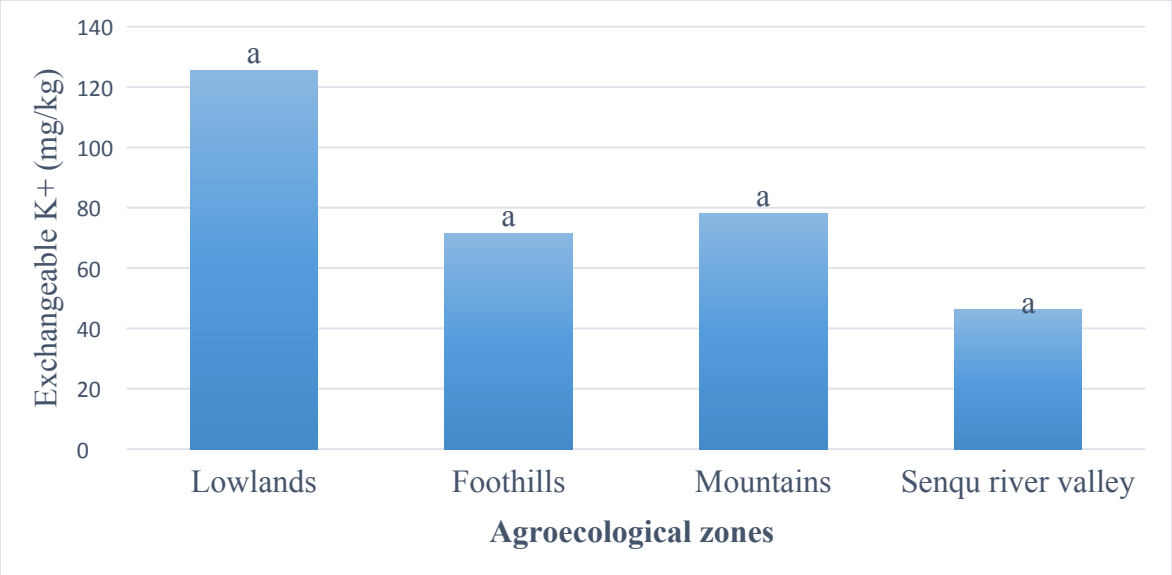


Figure 4.2. Average exchangeable K^+ content across agroecological zones of Lesotho.

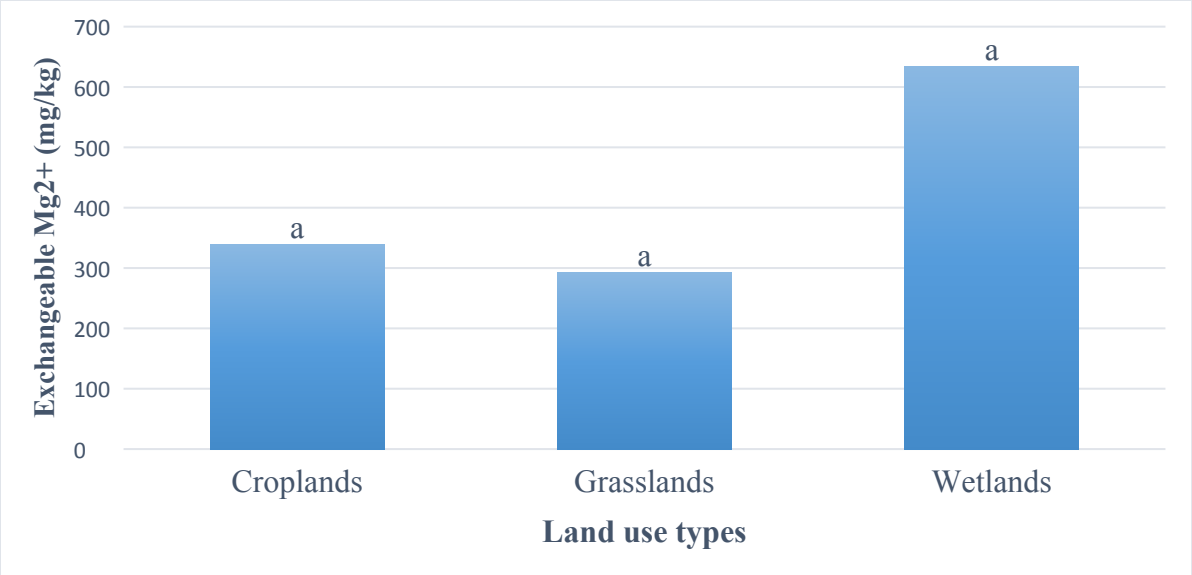


Figure 4.3. Average exchangeable Mg^{2+} content across three land use types.

Table 4.4. The comparison of the selected physicochemical properties of the study soils among (a) agroeco (b) land use types.

	Physical properties			
Agroecological zones	**~~Sand~~ ~~(%)~~**	**Silt (%)**	**Clay (%)**	**OC (%)**
Lowlands (n = 17)	~~51.95a~~	~~21.37a~~	26.68b	1.19a
Foothills (n = 8)	~~55.36a~~	~~19.08a~~	25.55b	2.40ab
Mountains (n = 38)	~~59.42a~~	~~23.65a~~	16.93a	6.39b
Senqu River Valley (n = 7)	~~56.04a~~	~~20.95a~~	23.01ab	0.95a

	Chemical properties					
Agroecological zones	**pH_{water}**	**pH_{KCl}**	**EC (dS/m)**	**CEC (cmol/kg)**	**Base saturation (%)**	**ESP (%)**
Lowlands (n = 17)	6.12ab	5.20a	0.06ab	27.97a	~~27.34a~~	0.40b

Foothills (n = 8)	6.11ab	5.10a	0.07ab	32.64ab	~~25.52a~~	0.14a	0.03a
Mountains (n = 38)	5.96a	5.12a	0.08b	54.56b	~~24.77a~~	0.17a	0.05a
Senqu River Valley (n = 7)	6.53b	5.86b	0.04a	24.78a	~~15.81a~~	0.12ab	0.03ab

Macronutrients

Agroecological zones	**Total N (%)**	**~~Available P (mg/kg)~~**	**Exchangeable Ca^2 (mg/kg)**	**Exchangeable Mg^2 (mg/kg)**	**Exchangeable K (mg/kg)**
Lowlands (n = 17)	0.10a	~~0.25a~~	786.21a	~~537.78a~~	~~125.65a~~
Foothills (n = 8)	0.16ab	~~0.21a~~	1251.55ab	~~497.74a~~	~~71.59a~~
Mountains (n = 38)	0.30b	~~0.16a~~	1636.77b	~~379.64a~~	~~78.22a~~
Senqu River Valley (n = 7)	0.06a	~~0.30a~~	625.36ab	~~117.93a~~	~~46.38a~~

Micronutrients and trace elements

Agroecological zones	**Exchangeable Na (mg/kg)**	**Extractable Fe (mg/kg)**	**Extractable Mn (mg/kg)**	**Extractable Cu (mg/kg)**	**Extractable Zn (mg/kg)**

Lowlands (n = 17)	31.11a	2.96a	3.37a	2.71a	2.24ab
Foothills (n = 8)	13.12a	7.27a	3.65a	5.29a	3.34ab
Mountains (n = 38)	21.80a	4.16a	2.97a	3.42a	3.51b
Senqu River Valley (n = 7)	6.82a	2.26a	3.12a	2.78a	1.55a

Physical properties

Land use types	**Sand (%)**	**Silt (%)**	**Clay (%)**	**OC (%)**	**Bulk density (kg/m³)**
Croplands (n = 16)	50.49a	21.88a	27.64b	1.49a	1791.54b
Grasslands (n = 33)	62.45b	21.29a	16.26a	4.55b	1556.03a
Wetlands (n = 21)	52.74ab	24.22a	23.04ab	5.47b	1483.49a

Chemical properties

Land use types	**pH_{water}**	**pH_{KCl}**	**EC (dS/m)**	**CEC (cmol/kg)**	**Base saturation (%)**	**ESP (%)**	**SAR**
Croplands (n = 16)	6.16a	5.14a	0.04a	38.55ab	19.93a	0.14ab	0.04ab

Grasslands (n = 33)	~~6.05a~~	~~5.22a~~	0.07ab	36.29a	22.98a	0.12a	0.03a
Wetlands (n = 21)	~~6.05a~~	~~5.25a~~	0.09b	55.67b	30.64a	0.43b	0.10b

Macronutrients

Land use types	**Total N (%)**	**Available P (mg/kg)**	**Exchangeable Ca² (mg/kg)**	**Exchangeable Mg² (mg/kg)**	**Exchangeable K (mg/kg)**
Croplands (n = 16)	0.08a	0.31a	1121.36a	~~338.87a~~	~~85.17a~~
Grasslands (n = 33)	0.22ab	0.19a	1075.78ab	~~292.05a~~	~~88.72a~~
Wetlands (n = 21)	0.31b	0.12a	1738.58ab	~~634.12a~~	~~81.70a~~

Micronutrients and trace elements

Land use types	**Exchangeable Na (mg/kg)**	**Extractable Fe (mg/kg)**	**Extractable Mn (mg/kg)**	**Extractable Cu (mg/kg)**	**Extractable Zn (mg/kg)**
Croplands (n = 16)	13.05a	1.85a	2.57a	3.15ab	~~2.40a~~
Grasslands (n = 33)	9.37a	3.91a	3.73a	2.67a	~~3.01a~~
Wetlands (n = 21)	47.24b	5.89b	2.72a	4.73b	~~3.39a~~

Average values followed by the same letter within the agroecological zones or within the land use types are not significantly different. Letters are based on Dunn's pairwise comparisons with Bonferroni–adjusted significance probability < 0.05).

Table 4.5. The comparison of the selected physicochemical properties of the study soils among (a) agroecological zones and (b) land use types interaction.

		Physical properties				
Agroecological zones	**Land use types**	**Sand (%)**	**~~Silt~~ (%)**	**Clay (%)**	**OC (%)**	**Bulk density (kg/m³)**
	Croplands (n = 3)	49.06ab	~~18.89a~~	32.05b	1.02abc	1814.68abc
Foothills	Grasslands (n = 3)	63.28ab	~~18.00a~~	18.72ab	3.26abc	1592.54abc
	Wetlands (n = 2)	52.95ab	~~21.00a~~	26.05ab	3.20abc	1639.56abc
	Croplands (n = 4)	60.45ab	~~16.83a~~	22.72ab	0.71a	1809.88bc
Lowlands	Grasslands (n = 6)	61.84ab	~~16.00a~~	22.16ab	0.48a	1818.06c
	Wetlands (n = 7)	38.61a	~~28.57a~~	32.82b	2.07abc	1617.88abc
Mountains	Croplands (n = 7)	46.71ab	~~25.90a~~	27.39b	2.12abc	1788.70c
	Grasslands (n = 20)	63.45b	~~23.47a~~	13.09a	6.71bc	1451.04ab

	Wetlands (n = 11)	60.19ab	~~22.55a~~	17.27ab	8.53c	1362.74a	
	Croplands (n = 2)	45.95ab	~~22.33a~~	31.72ab	1.53abc	1730.07abc	
Senqu River Valley	Grasslands (n = 4)	57.78ab	~~20.83a~~	21.39ab	0.87ab	1660.53abc	
	Wetlands (n = 1)	69.28ab	~~18.67a~~	12.05ab	0.13abc	1559.03abc	

Chemical properties

Agroecological zones	**Land use types**	**pH_{water}**	**pH_{KCl}**	**EC (dS/m)**	**CEC (cmol/kg)**	**~~Base saturation (%)~~**	**ESP (%)**	**SAR**
	Croplands (n = 3)	5.56a	4.53a	0.04a	22.32ab	~~12.46a~~	0.06ab	0.02ab
Foothills	Grasslands (n = 3)	6.53a	5.43ab	0.06a	39.18ab	~~30.85a~~	0.07ab	0.01ab
	Wetlands (n = 2)	6.29a	5.45ab	0.10a	38.33ab	~~37.13a~~	0.36ab	0.07ab
	Croplands (n = 4)	6.02a	4.90a	0.03a	23.22ab	~~15.32a~~	0.23ab	0.06ab
Lowlands	Grasslands (n = 6)	6.04a	5.18ab	0.04a	21.13a	~~12.79a~~	0.19ab	0.05ab
	Wetlands (n = 7)	6.24a	5.39ab	0.09a	36.56ab	~~46.67a~~	0.68b	0.13b
Mountains	Croplands (n = 7)	6.40a	5.37ab	0.05a	60.15ab	~~25.44a~~	0.12ab	0.04ab

	Grasslands (n = 20)	5.89b	5.09ab	0.08a	41.64ab	~~26.50a~~	0.11a	0.03a
	Wetlands (n = 11)	5.82ab	5.04ab	0.09a	74.48b	~~21.18a~~	0.30ab	0.09ab
	Croplands (n = 2)	6.47ab	5.72ab	0.04a	17.91ab	~~21.04a~~	0.12ab	0.02ab
Senqu River Valley	Grasslands (n = 4)	6.46ab	5.82b	0.04b	30.11ab	~~14.75a~~	0.11ab	0.03ab
	Wetlands (n = 1)	6.89ab	6.29ab	0.05a	17.23ab	~~9.55a~~	0.14ab	0.04ab

Macronutrients

Agroecological zones	**Land use types**	**Total N (%)**	**Available P (mg/kg)**	**Exchangeable Ca^2 (mg/kg)**	**~~Exchangeable Mg^2 (mg/kg)~~**	**~~Exchangeable K (mg/kg)~~**
	Croplands (n = 3)	0.12ab	~~0.30a~~	311.69ab	~~31.07a~~	~~57.67a~~
Foothills	Grasslands (n = 3)	0.14ab	~~0.21a~~	1511.75ab	~~849.63a~~	~~87.98a~~
	Wetlands (n = 2)	0.23ab	~~0.06a~~	2271.04ab	~~669.91a~~	~~67.88a~~
Lowlands	Croplands (n = 4)	0.05a	~~0.31a~~	495.60ab	~~238.10a~~	~~108.96a~~
	Grasslands (n = 6)	0.06a	~~0.29a~~	371.11a	~~105.83a~~	~~132.14a~~

	Wetlands (n = 7)	0.17ab	0.18a	1308.08ab	1079.26a	129.63a
	Croplands (n = 7)	0.09ab	0.18a	1982.33ab	572.63a	103.69a
Mountains	Grasslands (n = 20)	0.31ab	0.17a	1288.23ab	301.85a	78.75a
	Wetlands (n = 11)	0.44b	0.11a	2050.57b	398.28a	61.07a
	Croplands (n = 2)	0.04ab	0.78a	574.00ab	183.99a	14.06a
Senqu River Valley	Grasslands (n = 4)	0.06a	0.12a	743.56ab	104.19a	73.96a
	Wetlands (n = 1)	0.06ab	0.04a	255.25ab	40.78a	0.74a

Micronutrients and trace elements

Agroecological zones	**Land use types**	**Exchangeable Na (mg/kg)**	**Extractable Fe (mg/kg)**	**Extractable Mn (mg/kg)**	**Extractable Cu (mg/kg)**	**Extractable Zn (mg/kg)**
	Croplands (n = 4)	12.03ab	2.07ab	2.84a	2.12ab	2.83ab
Lowlands	Grasslands (n = 6)	10.31ab	1.00a	5.82a	0.93a	1.60a
	Wetlands (n = 7)	59.84a	5.15ab	1.58a	4.57ab	2.45ab
Foothills	Croplands (n = 3)	3.64ab	1.12ab	1.97a	4.71ab	2.75ab

	Grasslands (n = 3)	8.32ab	14.39ab	~~6.33a~~	5.04ab	3.03ab
	Wetlands (n = 2)	34.55ab	5.81ab	~~2.15a~~	6.53ab	4.68ab
	Croplands (n = 7)	19.74ab	2.15ab	~~3.15a~~	3.11ab	2.38ab
Mountains	Grasslands (n = 20)	9.58b	3.45ab	~~3.14a~~	2.88ab	3.63b
	Wetlands (n = 11)	45.32ab	6.73b	~~2.56a~~	4.60b	4.00ab
	Croplands (n = 2)	5.77ab	1.44ab	~~0.91a~~	3.02ab	1.13ab
Senqu River Valley	Grasslands (n = 4)	7.67ab	2.73ab	~~1.59a~~	2.46ab	2.01ab
	Wetlands (n = 1)	5.53ab	2.00ab	~~13.66a~~	3.58ab	0.54ab

Average values followed by the same letter within the agroecological zones or within the land use types are not significantly different. Letters are based on Dunn's pairwise comparisons with Bonferroni–adjusted significance probability < 0.05).

Table 4.6. The ranges and average values of the carbon to nutrient ratios of the soil samples obtained from the wetlands, grasslands and croplands land use types across the Lowlands, Foothills, Mountains and Senqu river valley zones of Lesotho, *together with the coefficient of variation (% CV) given in parentheses*.

Carbon to nutrient ratio	**Minimum**	**Maximum**	**Average (CV)**	**Variability class**

C:N	2.40	143.19	24.02 (100)	Extremely high
C:P	2,128.73	10,153,161.67	626,675.77 (210)	Extremely high
N:P	350.15	585,490.20	32,840.81 (228)	Extremely high

Variability class is based on the coefficient of variation (CV, %): Low = <20, Moderately low = 20–34, Moderate = 35–44, Moderately high = 45–54, High = 55–74, Very high = 75–94, Extremely high = ≥95.

Table 4.7. The ranges and average values of carbon to nutrient of the study soil samples pooled by (a) agroecological zones and (b) land use types, *together with the coefficient of variation (% CV) given in parentheses*.

Carbon to nutrient ratios				
Agroecological zones		**C:N**	**C:P**	**N:P**
Lowlands	Minimum	2.59	2128.73	787.29
	Maximum	59.72	2605701.63	175555.56
	Average(CV)	13.79(101)	230228.06(271))	18478.52(230)
	Variability Class	Extremely high	Extremely high	Extremely high
Foothills	Minimum	5.52	9729.22	552.92
	Maximum	37.8	2647914.77	157777.78
	Average(CV)	19.37(50)	612907.91(146)	33189.65(154)
	Variability Class	Moderately high	Extremely high	Extremely high
Mountains	Minimum	3.61	17152.08	1866.42
	Maximum	143.19	10153161.67	585490.20
	Average(CV)	29.71(95)	904415.40(181)	43598.48(216)
	Variability Class	Extremely high	Extremely high	Extremely high
Senqu river valley	Minimum	2.4	6116.81	350.15
	Maximum	65.62	297355.89	33523.81
	Average(CV)	23.26(100)	97482.6634(111)	8923.22(135)

	Variability Class	Extremely high	Extremely high	Extremely high
Land use types		**C:N**	**C:P**	**N:P**
Croplands	Minimum	2.59	2128.73	350.15
	Maximum	65.62	588940.35	46666.67
	Average(CV	22.48(87)	127596.1(124)	8581.78(137)
	Variability Class	Very high	Extremely high	Extremely high
Grasslands	Minimum	3.36	3166.12	1091.8
	Maximum	143.19	1740789.24	68761.9
	Average(CV)	28.65(107)	429779.7(103)	18515.85(94)
	Variability Class	Extremely high	Extremely high	Very high
Wetlands	Minimum	2.4	9969.87	2697.88
	Maximum	54.64	10153161.7	585490.2
	Average(CV)	17.9043(61)	1316335(167)	73834.54(172)
	Variability Class	High	Extremely high	Extremely high

Variability class is based on the coefficient of variation (CV, %): Low = <20, Moderately low = 20–34, Moderate = 35–44, Moderately high = 45–54, High = 55–74, Very high = 75–94, Extremely high = ≥95.

Table 4.7. Summary of the Kruskal–Wallies H test results showing the significance levels for the variation in soil carbon to nutrient ration as a function of agroecological zones and land use types in Lesotho.

	Source of variation		
Carbon to nutrient ratios	**LUT (df = 2)**	**AEZ (df = 3)**	**AEZ × LUT (df = 11)**
C:N	0.49877^{NS}	0.00653^{**}	0.03827^{*}

C:P	0.00173**	0.00003****	0.00004****
N:P	0.00041***	0.00124**	0.00024***

Superscripts show the significance level of the Kruskal–Wallis test: NS = not significant (P > 0.10); † = 0.05 ≤ P < 0.10; * = P ≤ 0.05; ** = P ≤ 0.01; *** = P ≤ 0.001; **** = P ≤ 0.0001.

Table 4.8. The comparison of the carbon to nutrient ratios of the study soils among (a) agroecological zones types.

	Carbon to nutrient ratios		
Agroecological zones	**C:N**	**C:P**	**N:P**
Lowlands (n = 17)	13.79a	230,228a	18,479a
Foothills (n = 8)	19.37ab	612,908ab	33,190ab
Mountains (n = 38)	29.71b	904,415b	43,598b
Senqu River Valley (n = 7)	23.26ab	97,483a	8,923ab

	Carbon to nutrient ratios		
Land use types	**C:N**	**C:P**	
Croplands (n = 16)	~~22.48a~~	28.65a	
Grasslands (n = 33)	~~127,596a~~	429,780b	1,
Wetlands (n = 21)	~~8,582a~~	18,516ab	

Average values followed by the same letter within the agroecological zones or within the land use types are not significantly different. Letters are based on Dunn's pairwise comparisons with Bonferroni–adjusted significance probability < 0.05.

Table 4.9. The comparison of the carbon to nutrient ratios of the study soils among (a) agroecological zone and (b) land use type interaction.

Carbon to nutrient ratios

Agroecological zone	Land use type	C:N	C:P	N:P
Lowlands	Croplands (n = 4)	17.16a	53,377.75ab	12,694.24ab
	Grasslands (n = 6)	10.63a	23,688.68a	2,544.54a
	Wetlands (n = 7)	14.59a	508,319.14abc	35,441.52ab
Foothills	Croplands (n = 3)	15.31a	121,863.81abc	10,846.48ab
	Grasslands (n = 3)	27.45a	574,416.86abc	19,526.09ab
	Wetlands (n = 2)	13.34a	1,407,210.63abc	87,199.73ab
Mountains	Croplands (n = 7)	22.93a	193,913.78abc	7,428.54ab
	Grasslands (n = 20)	36.19b	588,487.33bc	24,613.14ab
	Wetlands (n = 11)	22.25a	1,930,967.48c	101,134.52b

	Croplands (n = 2)	42.31a	52,519.01abc	996.19ab
Senqu River Valley	Grasslands (n = 4)	18.95a	136,899.89abc	11,228.65ab
	Wetlands (n = 1)	2.40a	29,741.07abc	15,555.56ab

Average values followed by the same letter within the agroecological zones or within the land use types are not significantly different. Letters are based on Dunn's pairwise comparisons with Bonferroni–adjusted significance probability < 0.05.

Appendix A1. Summary of Dunn's Post–Hoc Pairwise Comparisons with Bonferroni correction for agroecological zones on soil physicochemical properties and carbon to nutrient ratios following the Kruskal–Wallis H test results.

Physical properties					
	Sand	**Silt**	**Clay**	**OC**	**Bulk density**
Kruskal–Wallis H (df = 3)	3.246	7.660	16.800	33.131	18.289
Probability	0.35523	0.05358	0.00078	0.00000	0.00000
Dunn's test results: Probability level for pairwise mean comparisons					
L vs F	—	—	1.00000	1.00000	1.00000
L vs M	—	—	**0.00374***	**0.00000***	**0.00042***
L vs SRV	—	—	1.0000	1.00000	1.00000

F vs M	—	—	**0.02796***	0.11571	0.17377		
F vs SRV	—	—	1.00000	1.00000	1.00000		
M vs SRV	—	—	0.42068	**0.00181***	0.35551		
Chemical properties							
	pH_{water}	**pH_{KCl}**	**EC**	**CEC**	**Base saturation**	**ESP**	**SAR**
Kruskal–Wallis H (df = 3)	11.206	13.127	10.634	15.722	1.391	15.788	14.544
Probability	0.01066	0.00437	0.01388	0.00129	0.70763	0.00125	0.00225
Dunn's test results: Probability level for pairwise mean comparisons							
L vs F	1.00000	1.00000	1.00000	1.00000	—	**0.04727***	**0.01789***
L vs M	1.00000	1.00000	0.10936	**0.00806***	—	**0.00096***	**0.00628***
L vs SRV	0.11676	**0.01054***	1.00000	1.00000	—	0.08504	0.05456
F vs M	1.00000	1.00000	1.00000	0.37110	—	1.00000	1.00000
F vs SRV	0.59390	**0.04649***	0.84726	1.00000	—	1.00000	1.00000
M vs SRV	**0.00660***	**0.00217***	**0.04037***	**0.03059***	—	1.00000	1.00000
Macronutrients							

	Total N	Available P	Exchangeable Ca²	Exchangeable Mg²	Exchangeable K
Kruskal–Wallis H (df = 3)	23.988	2.367	12.183	3.821	7.730
Probability	0.00003	0.49989	0.00678	0.28145	0.05194
Dunn's test results: Probability level for pairwise mean comparisons					
L vs F	1.00000	—	1.00000	—	—
L vs M	**0.00067***	—	**0.01936***	—	—
L vs SRV	1.00000	—	1.00000	—	—
F vs M	0.27614	—	1.00000	—	—
F vs SRV	0.69875	—	1.00000	—	—
M vs SRV	**0.00141***	—	0.08434	—	—
Micronutrients and trace elements					
	Exchangeable Na	**Extractable Fe**	**Extractable Mn**	**Extractable Cu**	**Extractable Zn**
Kruskal–Wallis H (df = 3)	7.107	6.264	0.368	5.823	12.909
Probability	0.06855	0.09946	0.94683	0.12056	0.00484
Dunn's test results: Probability level for pairwise mean comparisons					
L vs F	—	—	—	—	0.77579

L vs M	—	—	—	—	0.06669
L vs SRV	—	—	—	—	1.00000
F vs M	—	—	—	—	1.00000
F vs SRV	—	—	—	—	0.17998
M vs SRV	—	—	—	—	**0.01902***

Carbon to nutrient ratios			
	C:N	**C:P**	**N:P**
Kruskal–Wallis H (df = 3)	12.266	23.554	15.811
Probability	0.00653	0.00003	0.00124
Dunn's test results: Probability level for pairwise mean comparisons			
L vs F	0.62770	0.24103	0.38394
L vs M	**0.00337***	**0.00009***	**0.00314***
L vs SRV	1.00000	1.00000	1.00000
F vs M	1.00000	1.00000	1.00000
F vs SRV	1.00000	0.54580	0.60039
M vs SRV	1.00000	**0.01315***	0.05609

H is the Kruskal–Wallis H statistic for the agroecological zones (df = 3) and Probability is its asymptotic significance. Pairwise comparisons show the Bonferroni–adjusted probability of Dunn's test for each of the six agroecological zone pairs; they are shown only where the Kruskal–Wallis test is significant, and a dash (—) indicates that pairwise comparisons were not performed. Bold font and an asterisk (*) indicates adjusted probability < 0.05. L = Lowlands (n = 17), F = Foothills (n = 8), M = Mountains (n = 38), and SRV = Senqu River Valley (n = 7).

Appendix A2. Summary of Dunn's Post–Hoc Pairwise Comparisons with Bonferroni correction for land use types on soil physicochemical properties and carbon to nutrient ratios following the Kruskal–Wallis H test results.

Physical properties							
	Sand	**Silt**	**Clay**	**OC**	**Bulk density**		
Kruskal–Wallis H (df = 3)	14.731	0.899	18.494	11.307	20.106		
Probability	0.00063	0.63781	0.00010	0.00350	0.00004		
Dunn's test results: Probability level for pairwise mean comparisons							
C vs G	**0.00069***	—	**0.00006***	**0.03033***	**0.00102***		
C vs W	0.45744	—	0.10662	**0.00298***	**0.00004***		
G vs W	0.06114	—	0.09772	0.80450	0.64007		
Chemical properties							
	pH_{water}	**pH_{KCl}**	**EC**	**CEC**	**Base saturation**	**ESP**	**SAR**
Kruskal–Wallis H (df = 3)	1.883	0.225	15.616	6.815	1.863	8.792	13.996
Probability	0.39005	0.89340	0.00023	0.03312	0.39397	0.01233	0.00091
Dunn's test results: Probability level for pairwise mean comparisons							

C vs G	—	—	0.05354	1.00000	—	1.00000	1.00000
C vs W	—	—	**0.00013***	0.08038	—	0.20668	0.06360
G vs W	—	—	0.06889	**0.04035***	—	**0.00973***	**0.00062***

Macronutrients

	Total N	**Available P**	**Exchangeable Ca²**	**Exchangeable Mg²**	**Exchangeable K**
Kruskal–Wallis H (df = 3)	16.914	1.392	6.855	5.870	0.269
Probability	0.00021	0.49868	0.03247	0.05313	0.87429
Dunn's test results: Probability level for pairwise mean comparisons					
C vs G	0.05585	—	1.00000	—	—
C vs W	**0.00012***	—	**0.04942***	—	—
G vs W	0.06215	—	0.07884	—	—

Micronutrients and trace elements

	Exchangeable Na	**Extractable Fe**	**Extractable Mn**	**Extractable Cu**	**Extractable Zn**

Kruskal–Wallis H (df = 3)	14.868	15.050	0.619	9.980	2.315
Probability	0.00059	0.00054	0.73372	0.00681	0.31421
Dunn's test results: Probability level for pairwise mean comparisons					
C vs G	1.00000	0.43442	—	0.78919	—
C vs W	**0.01805***	**0.00058***	—	0.30933	—
G vs W	**0.00057***	**0.01356***	—	**0.00475***	—

Carbon to nutrient ratios

	C:N	**C:P**	**N:P**
Kruskal–Wallis H (df = 3)	1.391	12.723	15.599
Probability	0.49877	0.00173	0.00041
Dunn's test results: Probability level for pairwise mean comparisons			
C vs G	—	**0.04489***	0.08576
C vs W	—	**0.00114***	**0.00025***
G vs W	—	0.34940	0.06582

H is the Kruskal–Wallis H statistic for the agroecological zones (df = 3) and probability is its asymptotic significance. Pairwise comparisons show the Bonferroni–adjusted probability of Dunn's test for each of the three land use type pairs; they

are shown only where the Kruskal–Wallis test is significant, and a dash (—) indicates that pairwise comparisons were not performed. Bold font and an asterisk (*) indicates adjusted probability < 0.05. C = Croplands (n = 16), G = Grasslands (n = 33), W = Wetlands (n = 21).

Appendix A3. Summary of Dunn's Post–Hoc Pairwise Comparisons with Bonferroni correction for agroecological zone and land use type interaction on soil physicochemical properties and carbon to nutrient ratios following the Kruskal–Wallis H test results.

Physical properties					
	Sand	**Silt**	**Clay**	**OC**	**Bulk density**
Kruskal–Wallis H (df = 11)	27.583	15.725	38.740	48.572	41.990
Probability	0.00375	0.15166	0.00006	0.00000	0.00002
Dunn's test results: Probability level for pairwise mean comparisons					
FH–C vs FH–G	1.00000	—	1.00000	1.00000	1.00000
FH–C vs FH–W	1.00000	—	1.00000	1.00000	1.00000
FH–C vs LL–C	1.00000	—	1.00000	1.00000	1.00000
FH–C vs LL–G	1.00000	—	1.00000	1.00000	1.00000
FH–C vs LL–W	1.00000	—	1.00000	1.00000	1.00000
FH–C vs MT–C	1.00000	—	1.00000	1.00000	1.00000
FH–C vs MT–G	0.59490	—	**0.01881***	1.00000	0.77788
FH–C vs MT–W	1.00000	—	0.83195	0.25351	0.19826
FH–C vs SRV–C	1.00000	—	1.00000	1.00000	1.00000
FH–C vs SRV–G	1.00000	—	1.00000	1.00000	1.00000
FH–C vs SRV–W	1.00000	—	1.00000	1.00000	1.00000
FH–G vs FH–W	1.00000	—	1.00000	1.00000	1.00000

FH–G vs LL–C	1.00000	—	1.00000	1.00000	1.00000
FH–G vs LL–G	1.00000	—	1.00000	1.00000	1.00000
FH–G vs LL–W	1.00000	—	1.00000	1.00000	1.00000
FH–G vs MT–C	1.00000	—	1.00000	1.00000	1.00000
FH–G vs MT–G	1.00000	—	1.00000	1.00000	1.00000
FH–G vs MT–W	1.00000	—	1.00000	1.00000	1.00000
FH–G vs SRV–C	1.00000	—	1.00000	1.00000	1.00000
FH–G vs SRV–G	1.00000	—	1.00000	1.00000	1.00000
FH–G vs SRV–W	1.00000	—	1.00000	1.00000	1.00000
FH–W vs LL–C	1.00000	—	1.00000	1.00000	1.00000
FH–W vs LL–G	1.00000	—	1.00000	1.00000	1.00000
FH–W vs LL–W	1.00000	—	1.00000	1.00000	1.00000
FH–W vs MT–C	1.00000	—	1.00000	1.00000	1.00000
FH–W vs MT–G	1.00000	—	1.00000	1.00000	1.00000
FH–W vs MT–W	1.00000	—	1.00000	1.00000	1.00000
FH–W vs SRV–C	1.00000	—	1.00000	1.00000	1.00000
FH–W vs SRV–G	1.00000	—	1.00000	1.00000	1.00000
FH–W vs SRV–W	1.00000	—	1.00000	1.00000	1.00000
LL–C vs LL–G	1.00000	—	1.00000	1.00000	1.00000
LL–C vs LL–W	1.00000	—	1.00000	1.00000	1.00000

LL–C vs MT–C	1.00000	—	1.00000	1.00000	1.00000
LL–C vs MT–G	1.00000	—	0.81128	**0.03188***	0.08441
LL–C vs MT–W	1.00000	—	1.00000	**0.00412***	**0.01659***
LL–C vs SRV–C	1.00000	—	1.00000	1.00000	1.00000
LL–C vs SRV–G	1.00000	—	1.00000	1.00000	1.00000
LL–C vs SRV–W	1.00000	—	1.00000	1.00000	1.00000
LL–G vs LL–W	1.00000	—	1.00000	1.00000	1.00000
LL–G vs MT–C	1.00000	—	1.00000	1.00000	1.00000
LL–G vs MT–G	1.00000	—	0.33654	**0.00227***	**0.00995***
LL–G vs MT–W	1.00000	—	1.00000	**0.00023***	**0.00167***
LL–G vs SRV–C	1.00000	—	1.00000	1.00000	1.00000
LL–G vs SRV–G	1.00000	—	1.00000	1.00000	1.00000
LL–G vs SRV–W	1.00000	—	1.00000	1.00000	1.00000
LL–W vs MT–C	1.00000	—	1.00000	1.00000	1.00000
LL–W vs MT–G	**0.04004***	—	**0.00602***	1.00000	1.00000
LL–W vs MT–W	1.00000	—	1.00000	0.14626	1.00000
LL–W vs SRV–C	1.00000	—	1.00000	1.00000	1.00000
LL–W vs SRV–G	1.00000	—	1.00000	1.00000	1.00000
LL–W vs SRV–W	1.00000	—	1.00000	1.00000	1.00000
MT–C vs MT–G	0.06933	—	**0.00758***	0.71002	**0.02749***

MT–C vs MT–W	1.00000	—	1.00000	0.09147	**0.00459***	
MT–C vs SRV–C	1.00000	—	1.00000	1.00000	1.00000	
MT–C vs SRV–G	1.00000	—	1.00000	1.00000	1.00000	
MT–C vs SRV–W	1.00000	—	1.00000	1.00000	1.00000	
MT–G vs MT–W	1.00000	—	1.00000	1.00000	1.00000	
MT–G vs SRV–C	1.00000	—	0.47651	1.00000	1.00000	
MT–G vs SRV–G	1.00000	—	1.00000	0.27785	1.00000	
MT–G vs SRV–W	1.00000	—	1.00000	1.00000	1.00000	
MT–W vs SRV–C	1.00000	—	1.00000	1.00000	1.00000	
MT–W vs SRV–G	1.00000	—	1.00000	**0.04220***	1.00000	
MT–W vs SRV–W	1.00000	—	1.00000	0.61058	1.00000	
SRV–C vs SRV–G	1.00000	—	1.00000	1.00000	1.00000	
SRV–C vs SRV–W	1.00000	—	1.00000	1.00000	1.00000	
SRV–G vs SRV–W	1.00000	—	1.00000	1.00000	1.00000	

Chemical properties

	pH_{water}	**pH_{KCl}**	**EC**	**CEC**	**Base saturation**	**ESP**	**SAR**
Kruskal–Wallis H (df = 11)	27.275	23.028	29.508	28.224	13.674	26.020	26.795
Probability	0.00418	0.01752	0.00189	0.00299	0.25154	0.00644	0.00493
Dunn's test results: Probability level for pairwise mean comparisons							

FH–C vs FH–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–C vs FH–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–C vs LL–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–C vs LL–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–C vs LL–W	1.00000	1.00000	1.00000	1.00000	—	0.23762	0.30921
FH–C vs MT–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–C vs MT–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–C vs MT–W	1.00000	1.00000	1.00000	0.31821	—	1.00000	1.00000
FH–C vs SRV–C	1.00000	0.62819	1.00000	1.00000	—	1.00000	1.00000
FH–C vs SRV–G	1.00000	**0.01213***	1.00000	1.00000	—	1.00000	1.00000
FH–C vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–G vs FH–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–G vs LL–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	0.87460
FH–G vs LL–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–G vs LL–W	1.00000	1.00000	1.00000	1.00000	—	0.17462	0.09588
FH–G vs MT–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–G vs MT–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–G vs MT–W	0.59787	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–G vs SRV–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–G vs SRV–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000

FH–G vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–W vs LL–C	1.00000	1.00000	0.48715	1.00000	—	1.00000	1.00000
FH–W vs LL–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–W vs LL–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–W vs MT–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–W vs MT–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–W vs MT–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–W vs SRV–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–W vs SRV–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
FH–W vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–C vs LL–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–C vs LL–W	1.00000	1.00000	0.59710	1.00000	—	1.00000	1.00000
LL–C vs MT–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–C vs MT–G	1.00000	1.00000	0.24737	1.00000	—	1.00000	1.00000
LL–C vs MT–W	1.00000	1.00000	0.05455	0.07081	—	1.00000	1.00000
LL–C vs SRV–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–C vs SRV–G	1.00000	0.99008	1.00000	1.00000	—	1.00000	1.00000
LL–C vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–G vs LL–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–G vs MT–C	1.00000	1.00000	1.00000	0.90004	—	1.00000	1.00000

LL–G vs MT–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–G vs MT–W	1.00000	1.00000	0.27307	**0.01109***	—	1.00000	1.00000
LL–G vs SRV–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–G vs SRV–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–G vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–W vs MT–C	1.00000	1.00000	1.00000	1.00000	—	0.31981	0.74305
LL–W vs MT–G	1.00000	1.00000	1.00000	1.00000	—	**0.01772***	**0.01774***
LL–W vs MT–W	1.00000	1.00000	1.00000	1.00000	—	0.69465	1.00000
LL–W vs SRV–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
LL–W vs SRV–G	1.00000	1.00000	1.00000	1.00000	—	0.36922	0.64405
LL–W vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
MT–C vs MT–G	**0.04451***	1.00000	1.00000	1.00000	—	1.00000	1.00000
MT–C vs MT–W	0.22012	1.00000	0.82690	1.00000	—	1.00000	1.00000
MT–C vs SRV–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
MT–C vs SRV–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
MT–C vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
MT–G vs MT–W	1.00000	1.00000	1.00000	0.35500	—	1.00000	1.00000
MT–G vs SRV–C	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
MT–G vs SRV–G	0.78185	0.43286	1.00000	1.00000	—	1.00000	1.00000

MT–G vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
MT–W vs SRV–C	1.00000	1.00000	1.00000	0.24649	—	1.00000	1.00000
MT–W vs SRV–G	0.38905	0.27324	**0.02645***	0.74366	—	1.00000	1.00000
MT–W vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
SRV–C vs SRV–G	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
SRV–C vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000
SRV–G vs SRV–W	1.00000	1.00000	1.00000	1.00000	—	1.00000	1.00000

Macronutrients

	Total N	**Available P**	**Exchangeable Ca^2**	**Exchangeable Mg^2**	**Exchangeable K**
Kruskal–Wallis H (df = 11)	40.752	8.630	25.790	15.337	12.307
Probability	0.00003	0.65602	0.00697	0.16760	0.34099
Dunn's test results: Probability level for pairwise mean comparisons					
FH–C vs FH–G	1.00000	—	1.00000	—	—
FH–C vs FH–W	1.00000	—	1.00000	—	—
FH–C vs LL–C	1.00000	—	1.00000	—	—
FH–C vs LL–G	1.00000	—	1.00000	—	—
FH–C vs LL–W	1.00000	—	1.00000	—	—
FH–C vs MT–C	1.00000	—	1.00000	—	—
FH–C vs MT–G	1.00000	—	1.00000	—	—

FH–C vs MT–W	0.48671	—	0.40145	—	—
FH–C vs SRV–C	1.00000	—	1.00000	—	—
FH–C vs SRV–G	1.00000	—	1.00000	—	—
FH–C vs SRV–W	1.00000	—	1.00000	—	—
FH–G vs FH–W	1.00000	—	1.00000	—	—
FH–G vs LL–C	1.00000	—	1.00000	—	—
FH–G vs LL–G	1.00000	—	1.00000	—	—
FH–G vs LL–W	1.00000	—	1.00000	—	—
FH–G vs MT–C	1.00000	—	1.00000	—	—
FH–G vs MT–G	1.00000	—	1.00000	—	—
FH–G vs MT–W	1.00000	—	1.00000	—	—
FH–G vs SRV–C	1.00000	—	1.00000	—	—
FH–G vs SRV–G	1.00000	—	1.00000	—	—
FH–G vs SRV–W	1.00000	—	1.00000	—	—
FH–W vs LL–C	1.00000	—	1.00000	—	—
FH–W vs LL–G	1.00000	—	1.00000	—	—
FH–W vs LL–W	1.00000	—	1.00000	—	—
FH–W vs MT–C	1.00000	—	1.00000	—	—
FH–W vs MT–G	1.00000	—	1.00000	—	—
FH–W vs MT–W	1.00000	—	1.00000	—	—

FH–W vs SRV–C	1.00000	—	1.00000	—	—
FH–W vs SRV–G	1.00000	—	1.00000	—	—
FH–W vs SRV–W	1.00000	—	1.00000	—	—
LL–C vs LL–G	1.00000	—	1.00000	—	—
LL–C vs LL–W	1.00000	—	1.00000	—	—
LL–C vs MT–C	1.00000	—	1.00000	—	—
LL–C vs MT–G	0.24041	—	1.00000	—	—
LL–C vs MT–W	**0.00664***	—	0.27012	—	—
LL–C vs SRV–C	1.00000	—	1.00000	—	—
LL–C vs SRV–G	1.00000	—	1.00000	—	—
LL–C vs SRV–W	1.00000	—	1.00000	—	—
LL–G vs LL–W	1.00000	—	1.00000	—	—
LL–G vs MT–C	1.00000	—	0.46039	—	—
LL–G vs MT–G	0.20129	—	0.90893	—	—
LL–G vs MT–W	**0.00333***	—	**0.02089***	—	—
LL–G vs SRV–C	1.00000	—	1.00000	—	—
LL–G vs SRV–G	1.00000	—	1.00000	—	—
LL–G vs SRV–W	1.00000	—	1.00000	—	—
LL–W vs MT–C	1.00000	—	1.00000	—	—
LL–W vs MT–G	1.00000	—	1.00000	—	—

LL–W vs MT–W	1.00000	—	1.00000	—	—
LL–W vs SRV–C	1.00000	—	1.00000	—	—
LL–W vs SRV–G	1.00000	—	1.00000	—	—
LL–W vs SRV–W	1.00000	—	1.00000	—	—
MT–C vs MT–G	1.00000	—	1.00000	—	—
MT–C vs MT–W	0.05620	—	1.00000	—	—
MT–C vs SRV–C	1.00000	—	1.00000	—	—
MT–C vs SRV–G	1.00000	—	1.00000	—	—
MT–C vs SRV–W	1.00000	—	1.00000	—	—
MT–G vs MT–W	1.00000	—	1.00000	—	—
MT–G vs SRV–C	1.00000	—	1.00000	—	—
MT–G vs SRV–G	0.34240	—	1.00000	—	—
MT–G vs SRV–W	1.00000	—	1.00000	—	—
MT–W vs SRV–C	0.08750	—	1.00000	—	—
MT–W vs SRV–G	**0.01020***	—	1.00000	—	—
MT–W vs SRV–W	1.00000	—	1.00000	—	—
SRV–C vs SRV–G	1.00000	—	1.00000	—	—
SRV–C vs SRV–W	1.00000	—	1.00000	—	—
SRV–G vs SRV–W	1.00000	—	1.00000	—	—
Micronutrients and trace elements					

	Exchangeable Na	Extractable Fe	Extractable Mn	Extractable Cu	Extractable Zn
Kruskal–Wallis H (df = 11)	21.638	26.577	6.585	20.453	20.402
Probability	0.02733	0.00532	0.83160	0.03950	0.04013
Dunn's test results: Probability level for pairwise mean comparisons					
FH–C vs FH–G	1.00000	1.00000	—	1.00000	1.00000
FH–C vs FH–W	1.00000	1.00000	—	1.00000	1.00000
FH–C vs LL–C	1.00000	1.00000	—	1.00000	1.00000
FH–C vs LL–G	1.00000	1.00000	—	0.78942	1.00000
FH–C vs LL–W	0.15707	1.00000	—	1.00000	1.00000
FH–C vs MT–C	1.00000	1.00000	—	1.00000	1.00000
FH–C vs MT–G	1.00000	1.00000	—	1.00000	1.00000
FH–C vs MT–W	0.93204	0.26976	—	1.00000	1.00000
FH–C vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
FH–C vs SRV–G	1.00000	1.00000	—	1.00000	1.00000
FH–C vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
FH–G vs FH–W	1.00000	1.00000	—	1.00000	1.00000
FH–G vs LL–C	1.00000	1.00000	—	1.00000	1.00000
FH–G vs LL–G	1.00000	1.00000	—	1.00000	1.00000
FH–G vs LL–W	0.59589	1.00000	—	1.00000	1.00000

FH–G vs MT–C	1.00000	1.00000	—	1.00000	1.00000
FH–G vs MT–G	1.00000	1.00000	—	1.00000	1.00000
FH–G vs MT–W	1.00000	1.00000	—	1.00000	1.00000
FH–G vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
FH–G vs SRV–G	1.00000	1.00000	—	1.00000	1.00000
FH–G vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
FH–W vs LL–C	1.00000	1.00000	—	1.00000	1.00000
FH–W vs LL–G	1.00000	0.34959	—	0.10429	1.00000
FH–W vs LL–W	1.00000	1.00000	—	1.00000	1.00000
FH–W vs MT–C	1.00000	1.00000	—	1.00000	1.00000
FH–W vs MT–G	1.00000	1.00000	—	1.00000	1.00000
FH–W vs MT–W	1.00000	1.00000	—	1.00000	1.00000
FH–W vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
FH–W vs SRV–G	1.00000	1.00000	—	1.00000	1.00000
FH–W vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
LL–C vs LL–G	1.00000	1.00000	—	1.00000	1.00000
LL–C vs LL–W	1.00000	1.00000	—	1.00000	1.00000
LL–C vs MT–C	1.00000	1.00000	—	1.00000	1.00000
LL–C vs MT–G	1.00000	1.00000	—	1.00000	1.00000
LL–C vs MT–W	1.00000	1.00000	—	1.00000	1.00000

LL–C vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
LL–C vs SRV–G	1.00000	1.00000	—	1.00000	1.00000
LL–C vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
LL–G vs LL–W	1.00000	0.57639	—	0.08725	1.00000
LL–G vs MT–C	1.00000	1.00000	—	0.94862	1.00000
LL–G vs MT–G	1.00000	0.37679	—	1.00000	**0.00634***
LL–G vs MT–W	1.00000	**0.00845***	—	**0.00561***	0.55825
LL–G vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
LL–G vs SRV–G	1.00000	1.00000	—	1.00000	1.00000
LL–G vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
LL–W vs MT–C	1.00000	1.00000	—	1.00000	1.00000
LL–W vs MT–G	**0.01406***	1.00000	—	1.00000	1.00000
LL–W vs MT–W	1.00000	1.00000	—	1.00000	1.00000
LL–W vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
LL–W vs SRV–G	0.52232	1.00000	—	1.00000	1.00000
LL–W vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
MT–C vs MT–G	1.00000	1.00000	—	1.00000	1.00000
MT–C vs MT–W	1.00000	0.47545	—	1.00000	1.00000
MT–C vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
MT–C vs SRV–G	1.00000	1.00000	—	1.00000	1.00000

MT–C vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
MT–G vs MT–W	1.00000	1.00000	—	1.00000	1.00000
MT–G vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
MT–G vs SRV–G	1.00000	1.00000	—	1.00000	1.00000
MT–G vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
MT–W vs SRV–C	1.00000	1.00000	—	1.00000	1.00000
MT–W vs SRV–G	1.00000	1.00000	—	1.00000	1.00000
MT–W vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
SRV–C vs SRV–G	1.00000	1.00000	—	1.00000	1.00000
SRV–C vs SRV–W	1.00000	1.00000	—	1.00000	1.00000
SRV–G vs SRV–W	1.00000	1.00000	—	1.00000	1.00000

Carbon to nutrient ratios

	C:N	C:P	N:P
Kruskal–Wallis H (df = 11)	20.556	39.640	35.097
Probability	0.03827	0.00004	0.00024
Dunn's test results: Probability level for pairwise mean comparisons			
FH–C vs FH–G	1.00000	1.00000	1.00000
FH–C vs FH–W	1.00000	1.00000	1.00000
FH–C vs LL–C	1.00000	1.00000	1.00000
FH–C vs LL–G	1.00000	1.00000	1.00000

FH–C vs LL–W	1.00000	1.00000	1.00000
FH–C vs MT–C	1.00000	1.00000	1.00000
FH–C vs MT–G	1.00000	1.00000	1.00000
FH–C vs MT–W	1.00000	0.64166	1.00000
FH–C vs SRV–C	1.00000	1.00000	1.00000
FH–C vs SRV–G	1.00000	1.00000	1.00000
FH–C vs SRV–W	1.00000	1.00000	1.00000
FH–G vs FH–W	1.00000	1.00000	1.00000
FH–G vs LL–C	1.00000	1.00000	1.00000
FH–G vs LL–G	1.00000	0.99055	1.00000
FH–G vs LL–W	1.00000	1.00000	1.00000
FH–G vs MT–C	1.00000	1.00000	1.00000
FH–G vs MT–G	1.00000	1.00000	1.00000
FH–G vs MT–W	1.00000	1.00000	1.00000
FH–G vs SRV–C	1.00000	1.00000	1.00000
FH–G vs SRV–G	1.00000	1.00000	1.00000
FH–G vs SRV–W	1.00000	1.00000	1.00000
FH–W vs LL–C	1.00000	1.00000	1.00000
FH–W vs LL–G	1.00000	1.00000	0.80241
FH–W vs LL–W	1.00000	1.00000	1.00000

FH–W vs MT–C	1.00000	1.00000	1.00000
FH–W vs MT–G	1.00000	1.00000	1.00000
FH–W vs MT–W	1.00000	1.00000	1.00000
FH–W vs SRV–C	1.00000	1.00000	1.00000
FH–W vs SRV–G	1.00000	1.00000	1.00000
FH–W vs SRV–W	1.00000	1.00000	1.00000
LL–C vs LL–G	1.00000	1.00000	1.00000
LL–C vs LL–W	1.00000	1.00000	1.00000
LL–C vs MT–C	1.00000	1.00000	1.00000
LL–C vs MT–G	1.00000	0.27800	1.00000
LL–C vs MT–W	1.00000	**0.02103***	0.13690
LL–C vs SRV–C	1.00000	1.00000	1.00000
LL–C vs SRV–G	1.00000	1.00000	1.00000
LL–C vs SRV–W	1.00000	1.00000	1.00000
LL–G vs LL–W	1.00000	1.00000	1.00000
LL–G vs MT–C	1.00000	1.00000	1.00000
LL–G vs MT–G	**0.00303***	**0.01068***	0.08733
LL–G vs MT–W	1.00000	**0.00042***	**0.00078***
LL–G vs SRV–C	1.00000	1.00000	1.00000
LL–G vs SRV–G	1.00000	1.00000	1.00000

LL–G vs SRV–W	1.00000	1.00000	1.00000
LL–W vs MT–C	1.00000	1.00000	1.00000
LL–W vs MT–G	1.00000	1.00000	1.00000
LL–W vs MT–W	1.00000	0.73282	0.94716
LL–W vs SRV–C	1.00000	1.00000	1.00000
LL–W vs SRV–G	1.00000	1.00000	1.00000
LL–W vs SRV–W	1.00000	1.00000	1.00000
MT–C vs MT–G	1.00000	1.00000	1.00000
MT–C vs MT–W	1.00000	0.21250	0.05088
MT–C vs SRV–C	1.00000	1.00000	1.00000
MT–C vs SRV–G	1.00000	1.00000	1.00000
MT–C vs SRV–W	1.00000	1.00000	1.00000
MT–G vs MT–W	1.00000	1.00000	1.00000
MT–G vs SRV–C	1.00000	1.00000	0.81691
MT–G vs SRV–G	1.00000	1.00000	1.00000
MT–G vs SRV–W	1.00000	1.00000	1.00000
MT–W vs SRV–C	1.00000	0.45467	0.05112
MT–W vs SRV–G	1.00000	0.46463	0.62340
MT–W vs SRV–W	1.00000	1.00000	1.00000
SRV–C vs SRV–G	1.00000	1.00000	1.00000

SRV–C vs SRV–W	1.00000	1.00000	1.00000
SRV–G vs SRV–W	1.00000	1.00000	1.00000

H is the Kruskal–Wallis H statistic for the agroecological zone × land use type interaction (df = 11) and Probability is its asymptotic significance. Pairwise comparisons show the Bonferroni–adjusted probability of Dunn's test for each of the 66 pairs of AEZ × LUT groups; they are shown only where the Kruskal–Wallis test is significant, and a dash (—) indicates no significant differences among the pairs. Bold font and an asterisk (*) indicates adjusted probability < 0.05. FH = Foothills, LL = Lowlands, MT = Mountains, SRV = Senqu River Valley; C = Cropland, G = Grassland, W = Wetland.

Appendix B1. Spearman's rank correlation coefficients and unadjusted and Holm-adjusted probability values for pairwise correlations among soil properties and carbon to nutrient ratios for soil samples obtained from the wetlands, grasslands and croplands land use types across the Lowlands, Foothills, Mountains and Senqu river valley zones of Lesotho

Soil property 1	Soil property 2	Spearman's ρ	Raw p	Rank	Multiplier	Raw adjusted	Holm-adjusted p	Sig. raw Prob< 0.05	Sig. Holm prob < 0.05
C:N	C:P	0.547	0.000	1	300	0.000	0.000	Yes	Yes
C:N	OC	0.551	0.000	2	299	0.000	0.000	Yes	Yes
C:P	N:P	0.884	0.000	3	298	0.000	0.000	Yes	Yes
C:P	EC	0.522	0.000	4	297	0.000	0.000	Yes	Yes
C:P	Bulk density	-0.684	0.000	5	296	0.000	0.000	Yes	Yes
C:P	CEC	0.446	0.000	6	295	0.000	0.000	Yes	Yes
C:P	Fe	0.605	0.000	7	294	0.000	0.000	Yes	Yes
C:P	Zn	0.437	0.000	8	293	0.000	0.000	Yes	Yes
C:P	OC	0.831	0.000	9	292	0.000	0.000	Yes	Yes
C:P	N	0.607	0.000	10	291	0.000	0.000	Yes	Yes
C:P	P	-0.683	0.000	11	290	0.000	0.000	Yes	Yes
N:P	EC	0.450	0.000	12	289	0.000	0.000	Yes	Yes
N:P	Bulk density	-0.647	0.000	13	288	0.000	0.000	Yes	Yes
N:P	Fe	0.614	0.000	14	287	0.000	0.000	Yes	Yes

N:P	Zn	0.425	0.000	15	286	0.000	0.000	Yes	Yes
N:P	OC	0.676	0.000	16	285	0.000	0.000	Yes	Yes
N:P	N	0.705	0.000	17	284	0.000	0.000	Yes	Yes
N:P	P	-0.738	0.000	18	283	0.000	0.000	Yes	Yes
pHwater	pHKCl	0.860	0.000	19	282	0.000	0.000	Yes	Yes
EC	Bulk density	-0.524	0.000	20	281	0.000	0.000	Yes	Yes
EC	Silt	0.427	0.000	21	280	0.000	0.000	Yes	Yes
EC	Ca^{2+}	0.540	0.000	22	279	0.000	0.000	Yes	Yes
EC	Base saturation	0.568	0.000	23	278	0.000	0.000	Yes	Yes
EC	Fe	0.431	0.000	24	277	0.000	0.000	Yes	Yes
EC	Zn	0.444	0.000	25	276	0.000	0.000	Yes	Yes
EC	OC	0.609	0.000	26	275	0.000	0.000	Yes	Yes
EC	N	0.555	0.000	27	274	0.000	0.000	Yes	Yes
Bulk density	Clay	0.583	0.000	28	273	0.000	0.000	Yes	Yes
Bulk density	Fe	-0.573	0.000	29	272	0.000	0.000	Yes	Yes
Bulk density	Zn	-0.443	0.000	30	271	0.000	0.000	Yes	Yes

Bulk density	OC	-0.799	0.000	31	270	0.000	0.000	Yes	Yes
Bulk density	N	-0.722	0.000	32	269	0.000	0.000	Yes	Yes
Sand	Silt	-0.532	0.000	33	268	0.000	0.000	Yes	Yes
Sand	Clay	-0.770	0.000	34	267	0.000	0.000	Yes	Yes
Silt	Ca^{2+}	0.406	0.000	35	266	0.000	0.000	Yes	Yes
Silt	OC	0.446	0.000	36	265	0.000	0.000	Yes	Yes
Clay	OC	-0.473	0.000	37	264	0.000	0.000	Yes	Yes
Clay	N	-0.480	0.000	38	263	0.000	0.000	Yes	Yes
Ca^{2+}	Mg^{2+}	0.669	0.000	39	262	0.000	0.000	Yes	Yes
Ca^{2+}	K^{+}	0.451	0.000	40	261	0.000	0.000	Yes	Yes
Ca^{2+}	Na^{+}	0.648	0.000	41	260	0.000	0.000	Yes	Yes
Ca^{2+}	CEC	0.657	0.000	42	259	0.000	0.000	Yes	Yes
Ca^{2+}	Base saturation	0.748	0.000	43	258	0.000	0.000	Yes	Yes
Ca^{2+}	Fe	0.416	0.000	44	257	0.000	0.000	Yes	Yes
Ca^{2+}	Cu	0.474	0.000	45	256	0.000	0.000	Yes	Yes
Ca^{2+}	OC	0.517	0.000	46	255	0.000	0.000	Yes	Yes
Ca^{2+}	N	0.533	0.000	47	254	0.000	0.000	Yes	Yes
Mg^{2+}	K^{+}	0.461	0.000	48	253	0.000	0.000	Yes	Yes

Mg^{2+}	Na^{+}	0.792	0.000	49	252	0.000	0.000	Yes	Yes
Mg^{2+}	CEC	0.551	0.000	50	251	0.000	0.000	Yes	Yes
Mg^{2+}	Base saturation	0.638	0.000	51	250	0.000	0.000	Yes	Yes
Mg^{2+}	SAR	0.562	0.000	52	249	0.000	0.000	Yes	Yes
Mg^{2+}	ESP	0.572	0.000	53	248	0.000	0.000	Yes	Yes
Mg^{2+}	Cu	0.559	0.000	54	247	0.000	0.000	Yes	Yes
K^{+}	Na^{+}	0.559	0.000	55	246	0.000	0.000	Yes	Yes
K^{+}	Base saturation	0.564	0.000	56	245	0.000	0.000	Yes	Yes
K^{+}	SAR	0.446	0.000	57	244	0.000	0.000	Yes	Yes
K^{+}	ESP	0.524	0.000	58	243	0.000	0.000	Yes	Yes
Na^{+}	CEC	0.522	0.000	59	242	0.000	0.000	Yes	Yes
Na^{+}	Base saturation	0.550	0.000	60	241	0.000	0.000	Yes	Yes
Na^{+}	SAR	0.888	0.000	61	240	0.000	0.000	Yes	Yes
Na^{+}	ESP	0.841	0.000	62	239	0.000	0.000	Yes	Yes
Na^{+}	Cu	0.441	0.000	63	238	0.000	0.000	Yes	Yes
CEC	OC	0.449	0.000	64	237	0.000	0.000	Yes	Yes
Base saturation	ESP	0.560	0.000	65	236	0.000	0.000	Yes	Yes

Base saturation	Cu	0.408	0.000	66	235	0.000	0.000	Yes	Yes
SAR	ESP	0.877	0.000	67	234	0.000	0.000	Yes	Yes
Fe	Cu	0.418	0.000	68	233	0.000	0.000	Yes	Yes
Fe	Zn	0.680	0.000	69	232	0.000	0.000	Yes	Yes
Fe	OC	0.588	0.000	70	231	0.000	0.000	Yes	Yes
Fe	N	0.615	0.000	71	230	0.000	0.000	Yes	Yes
Zn	OC	0.558	0.000	72	229	0.000	0.000	Yes	Yes
Zn	N	0.545	0.000	73	228	0.000	0.000	Yes	Yes
OC	N	0.783	0.000	74	227	0.000	0.000	Yes	Yes
C:P	Silt	0.379	0.001	75	226	0.226	0.226	Yes	No
C:P	Ca^{2+}	0.405	0.001	76	225	0.225	0.226	Yes	No
N:P	pHwater	-0.380	0.001	77	224	0.224	0.226	Yes	No
N:P	Clay	-0.385	0.001	78	223	0.223	0.226	Yes	No
pHwater	OC	-0.381	0.001	79	222	0.222	0.226	Yes	No
pHwater	N	-0.397	0.001	80	221	0.221	0.226	Yes	No
EC	K^{+}	0.402	0.001	81	220	0.220	0.226	Yes	No
Silt	Mg^{2+}	0.386	0.001	82	219	0.219	0.226	Yes	No
Silt	N	0.374	0.001	83	218	0.218	0.226	Yes	No
Ca^{2+}	Zn	0.394	0.001	84	217	0.217	0.226	Yes	No

CEC	Cu	0.377	0.001	85	216	0.216	0.226	Yes	No
CEC	N	0.382	0.001	86	215	0.215	0.226	Yes	No
N:P	pHKCl	-0.357	0.002	87	214	0.428	0.428	Yes	No
EC	Na^{+}	0.358	0.002	88	213	0.426	0.428	Yes	No
Silt	Base saturation	0.365	0.002	89	212	0.424	0.428	Yes	No
C:P	Clay	-0.350	0.003	90	211	0.633	0.633	Yes	No
N:P	CEC	0.350	0.003	91	210	0.630	0.633	Yes	No
pHwater	Bulk density	0.351	0.003	92	209	0.627	0.633	Yes	No
pHwater	Clay	0.348	0.003	93	208	0.624	0.633	Yes	No
EC	Mg^{2+}	0.353	0.003	94	207	0.621	0.633	Yes	No
EC	Cu	0.345	0.003	95	206	0.618	0.633	Yes	No
Bulk density	Silt	-0.348	0.003	96	205	0.615	0.633	Yes	No
Sand	Mg^{2+}	-0.353	0.003	97	204	0.612	0.633	Yes	No
Ca^{2+}	ESP	0.355	0.003	98	203	0.609	0.633	Yes	No
CEC	Fe	0.351	0.003	99	202	0.606	0.633	Yes	No
C:P	pHwater	-0.341	0.004	100	201	0.804	0.804	Yes	No
N:P	Ca^{2+}	0.342	0.004	101	200	0.800	0.804	Yes	No
Clay	Zn	-0.344	0.004	102	199	0.796	0.804	Yes	No

C:P	Cu	0.333	0.005	103	198	0.990	0.990	Yes	No
Silt	CEC	0.336	0.005	104	197	0.985	0.990	Yes	No
Clay	Fe	-0.334	0.005	105	196	0.980	0.990	Yes	No
Cu	P	-0.334	0.005	106	195	0.975	0.990	Yes	No
C:N	ESP	-0.328	0.006	107	194	1.000	1.000	Yes	No
Sand	K^+	-0.328	0.006	108	193	1.000	1.000	Yes	No
pHwater	Fe	-0.320	0.007	109	192	1.000	1.000	Yes	No
Silt	K^+	0.319	0.007	110	191	1.000	1.000	Yes	No
Silt	Fe	0.316	0.008	111	190	1.000	1.000	Yes	No
Sand	Na^+	-0.311	0.009	112	189	1.000	1.000	Yes	No
Cu	Zn	0.312	0.009	113	188	1.000	1.000	Yes	No
N:P	Silt	0.305	0.010	114	187	1.000	1.000	Yes	No
N:P	Cu	0.306	0.010	115	186	1.000	1.000	Yes	No
Silt	Na^+	0.307	0.010	116	185	1.000	1.000	Yes	No
Fe	P	-0.305	0.010	117	184	1.000	1.000	Yes	No
C:N	SAR	-0.303	0.011	118	183	1.000	1.000	Yes	No
C:P	pHKCl	-0.303	0.011	119	182	1.000	1.000	Yes	No
CEC	SAR	0.302	0.011	120	181	1.000	1.000	Yes	No
Sand	Cu	-0.298	0.012	121	180	1.000	1.000	Yes	No
Ca^{2+}	SAR	0.298	0.012	122	179	1.000	1.000	Yes	No

C:N	Bulk density	-0.296	0.013	123	178	1.000	1.000	Yes	No
Base saturation	N	0.291	0.014	124	177	1.000	1.000	Yes	No
pHwater	Zn	-0.289	0.015	125	176	1.000	1.000	Yes	No
ESP	Cu	0.290	0.015	126	175	1.000	1.000	Yes	No
Silt	Cu	0.287	0.016	127	174	1.000	1.000	Yes	No
Sand	Base saturation	-0.285	0.017	128	173	1.000	1.000	Yes	No
Sand	ESP	-0.282	0.018	129	172	1.000	1.000	Yes	No
Na^{+}	Fe	0.280	0.019	130	171	1.000	1.000	Yes	No
Bulk density	Ca^{2+}	-0.278	0.020	131	170	1.000	1.000	Yes	No
Mg^{2+}	Fe	0.275	0.021	132	169	1.000	1.000	Yes	No
Base saturation	Zn	0.274	0.022	133	168	1.000	1.000	Yes	No
C:N	EC	0.272	0.023	134	167	1.000	1.000	Yes	No
Base saturation	Fe	0.271	0.023	135	166	1.000	1.000	Yes	No
pHwater	K^{+}	0.269	0.024	136	165	1.000	1.000	Yes	No
SAR	Cu	0.270	0.024	137	164	1.000	1.000	Yes	No

pHwater	Base saturation	0.266	0.026	138	163	1.000	1.000	Yes	No
CEC	Zn	0.263	0.028	139	162	1.000	1.000	Yes	No
C:N	CEC	0.260	0.030	140	161	1.000	1.000	Yes	No
pHKCl	N	-0.259	0.030	141	160	1.000	1.000	Yes	No
Clay	ESP	0.258	0.031	142	159	1.000	1.000	Yes	No
pHKCl	Clay	0.257	0.032	143	158	1.000	1.000	Yes	No
pHKCl	Base saturation	0.257	0.032	144	157	1.000	1.000	Yes	No
pHwater	Mg^{2+}	0.254	0.034	145	156	1.000	1.000	Yes	No
pHwater	Sand	-0.252	0.035	146	155	1.000	1.000	Yes	No
EC	Clay	-0.248	0.038	147	154	1.000	1.000	Yes	No
EC	ESP	0.248	0.039	148	153	1.000	1.000	Yes	No
Bulk density	CEC	-0.247	0.039	149	152	1.000	1.000	Yes	No
Base saturation	SAR	0.247	0.039	150	151	1.000	1.000	Yes	No
C:N	P	-0.245	0.041	151	150	1.000	1.000	Yes	No
Bulk density	Sand	-0.241	0.045	152	149	1.000	1.000	Yes	No
ESP	OC	-0.241	0.045	153	148	1.000	1.000	Yes	No

Base saturation	OC	0.239	0.047	154	147	1.000	1.000	Yes	No
Sand	Zn	0.236	0.049	155	146	1.000	1.000	Yes	No
Sand	SAR	-0.235	0.050	156	145	1.000	1.000	No	No
pHKCl	OC	-0.229	0.056	157	144	1.000	1.000	No	No
pHKCl	Mn	-0.227	0.059	158	143	1.000	1.000	No	No
Bulk density	P	0.226	0.060	159	142	1.000	1.000	No	No
C:N	Silt	0.225	0.061	160	141	1.000	1.000	No	No
Mn	P	-0.225	0.061	161	140	1.000	1.000	No	No
Clay	Mg^{2+}	0.219	0.068	162	139	1.000	1.000	No	No
OC	P	-0.219	0.068	163	138	1.000	1.000	No	No
Bulk density	ESP	0.213	0.076	164	137	1.000	1.000	No	No
pHKCl	Mg^{2+}	0.212	0.078	165	136	1.000	1.000	No	No
pHKCl	K^{+}	0.211	0.080	166	135	1.000	1.000	No	No
Mg^{2+}	Zn	0.209	0.082	167	134	1.000	1.000	No	No
SAR	OC	-0.207	0.086	168	133	1.000	1.000	No	No
Clay	SAR	0.205	0.089	169	132	1.000	1.000	No	No
C:P	ESP	-0.204	0.091	170	131	1.000	1.000	No	No
pHwater	ESP	0.203	0.092	171	130	1.000	1.000	No	No

K^+	Mn	-0.202	0.093	172	129	1.000	1.000	No	No
K^+	P	0.201	0.095	173	128	1.000	1.000	No	No
C:N	Mn	-0.200	0.096	174	127	1.000	1.000	No	No
EC	CEC	0.199	0.098	175	126	1.000	1.000	No	No
pHKCl	Ca^{2+}	0.194	0.108	176	125	1.000	1.000	No	No
pHKCl	P	0.192	0.111	177	124	1.000	1.000	No	No
pHKCl	Bulk density	0.190	0.115	178	123	1.000	1.000	No	No
Clay	Na^+	0.190	0.115	179	122	1.000	1.000	No	No
SAR	Mn	0.189	0.117	180	121	1.000	1.000	No	No
Na^+	N	0.187	0.121	181	120	1.000	1.000	No	No
K^+	CEC	0.185	0.126	182	119	1.000	1.000	No	No
pHwater	Cu	0.183	0.128	183	118	1.000	1.000	No	No
Clay	K^+	0.182	0.132	184	117	1.000	1.000	No	No
Sand	Ca^{2+}	-0.181	0.134	185	116	1.000	1.000	No	No
pHwater	Na^+	0.178	0.140	186	115	1.000	1.000	No	No
Clay	Cu	0.178	0.140	187	114	1.000	1.000	No	No
pHwater	Ca^{2+}	0.176	0.144	188	113	1.000	1.000	No	No
pHKCl	Sand	-0.173	0.151	189	112	1.000	1.000	No	No
CEC	P	-0.173	0.152	190	111	1.000	1.000	No	No

C:N	Fe	0.171	0.156	191	110	1.000	1.000	No	No
pHKCl	Fe	-0.172	0.156	192	109	1.000	1.000	No	No
N:P	Mn	0.171	0.157	193	108	1.000	1.000	No	No
Bulk density	SAR	0.168	0.164	194	107	1.000	1.000	No	No
C:P	Base saturation	0.166	0.170	195	106	1.000	1.000	No	No
C:N	N:P	0.163	0.178	196	105	1.000	1.000	No	No
K^+	Cu	0.163	0.178	197	104	1.000	1.000	No	No
pHwater	Mn	-0.162	0.180	198	103	1.000	1.000	No	No
K^+	Zn	0.162	0.180	199	102	1.000	1.000	No	No
C:N	Ca^{2+}	0.160	0.185	200	101	1.000	1.000	No	No
Fe	Mn	0.152	0.210	201	100	1.000	1.000	No	No
EC	SAR	0.151	0.213	202	99	1.000	1.000	No	No
Cu	OC	0.149	0.219	203	98	1.000	1.000	No	No
Clay	Mn	-0.149	0.220	204	97	1.000	1.000	No	No
ESP	P	0.148	0.221	205	96	1.000	1.000	No	No
Cu	N	0.148	0.222	206	95	1.000	1.000	No	No
C:N	Zn	0.147	0.223	207	94	1.000	1.000	No	No
Silt	SAR	0.147	0.224	208	93	1.000	1.000	No	No
Silt	Zn	0.146	0.229	209	92	1.000	1.000	No	No

Sand	N	0.145	0.231	210	91	1.000	1.000	No	No
Silt	ESP	0.137	0.257	211	90	1.000	1.000	No	No
C:P	SAR	-0.133	0.274	212	89	1.000	1.000	No	No
pHKCl	ESP	0.132	0.276	213	88	1.000	1.000	No	No
Base saturation	Mn	-0.131	0.280	214	87	1.000	1.000	No	No
pHKCl	Na^+	0.131	0.281	215	86	1.000	1.000	No	No
Na^+	Zn	0.130	0.283	216	85	1.000	1.000	No	No
N:P	Base saturation	0.129	0.287	217	84	1.000	1.000	No	No
C:N	Na^+	-0.126	0.298	218	83	1.000	1.000	No	No
Mn	Cu	0.124	0.306	219	82	1.000	1.000	No	No
SAR	Fe	0.124	0.307	220	81	1.000	1.000	No	No
C:P	Mg^{2+}	0.122	0.315	221	80	1.000	1.000	No	No
EC	Mn	-0.119	0.327	222	79	1.000	1.000	No	No
ESP	Fe	0.116	0.341	223	78	1.000	1.000	No	No
Mg^{2+}	OC	0.115	0.344	224	77	1.000	1.000	No	No
N:P	ESP	-0.115	0.345	225	76	1.000	1.000	No	No
EC	P	-0.114	0.346	226	75	1.000	1.000	No	No
pHwater	P	0.111	0.360	227	74	1.000	1.000	No	No
Sand	Fe	0.111	0.362	228	73	1.000	1.000	No	No

Mg^{2+}	N	0.110	0.364	229	72	1.000	1.000	No	No
Sand	Mn	0.110	0.366	230	71	1.000	1.000	No	No
N:P	Sand	0.108	0.372	231	70	1.000	1.000	No	No
Mn	OC	-0.108	0.372	232	69	1.000	1.000	No	No
CEC	Base saturation	0.108	0.373	233	68	1.000	1.000	No	No
Base saturation	P	0.107	0.379	234	67	1.000	1.000	No	No
N	P	-0.106	0.380	235	66	1.000	1.000	No	No
pHwater	SAR	0.104	0.391	236	65	1.000	1.000	No	No
Ca^{2+}	Mn	-0.103	0.396	237	64	1.000	1.000	No	No
pHKCl	Cu	0.102	0.401	238	63	1.000	1.000	No	No
pHKCl	Zn	-0.100	0.409	239	62	1.000	1.000	No	No
Sand	OC	0.100	0.409	240	61	1.000	1.000	No	No
Sand	CEC	-0.098	0.418	241	60	1.000	1.000	No	No
ESP	Mn	0.095	0.432	242	59	1.000	1.000	No	No
K^{+}	N	0.089	0.462	243	58	1.000	1.000	No	No
C:N	Clay	-0.085	0.485	244	57	1.000	1.000	No	No
Na^{+}	Mn	0.082	0.499	245	56	1.000	1.000	No	No
C:N	Sand	-0.082	0.502	246	55	1.000	1.000	No	No
N:P	Na^{+}	0.082	0.502	247	54	1.000	1.000	No	No

Silt	P	-0.081	0.506	248	53	1.000	1.000	No	No
Clay	Base saturation	0.080	0.508	249	52	1.000	1.000	No	No
Bulk density	K^+	0.080	0.509	250	51	1.000	1.000	No	No
Bulk density	Cu	-0.079	0.516	251	50	1.000	1.000	No	No
Bulk density	Base saturation	-0.077	0.524	252	49	1.000	1.000	No	No
C:N	Cu	0.072	0.551	253	48	1.000	1.000	No	No
EC	Sand	-0.070	0.567	254	47	1.000	1.000	No	No
Clay	CEC	-0.067	0.579	255	46	1.000	1.000	No	No
C:N	pHwater	-0.065	0.591	256	45	1.000	1.000	No	No
C:P	Na^+	0.065	0.592	257	44	1.000	1.000	No	No
Clay	P	0.065	0.593	258	43	1.000	1.000	No	No
Bulk density	Mn	-0.063	0.604	259	42	1.000	1.000	No	No
pHKCl	Silt	-0.062	0.608	260	41	1.000	1.000	No	No
Bulk density	Mg^{2+}	0.062	0.609	261	40	1.000	1.000	No	No
Mn	Zn	-0.060	0.619	262	39	1.000	1.000	No	No
Silt	Clay	-0.058	0.635	263	38	1.000	1.000	No	No

SAR	P	0.057	0.639	264	37	1.000	1.000	No	No
C:N	Mg^{2+}	0.056	0.644	265	36	1.000	1.000	No	No
N:P	K^{+}	-0.056	0.645	266	35	1.000	1.000	No	No
Zn	P	-0.055	0.650	267	34	1.000	1.000	No	No
Bulk density	Na^{+}	0.049	0.686	268	33	1.000	1.000	No	No
SAR	Zn	-0.049	0.686	269	32	1.000	1.000	No	No
N:P	SAR	-0.049	0.688	270	31	1.000	1.000	No	No
pHwater	CEC	-0.049	0.688	271	30	1.000	1.000	No	No
Na^{+}	OC	0.048	0.692	272	29	1.000	1.000	No	No
pHKCl	EC	0.047	0.699	273	28	1.000	1.000	No	No
Na^{+}	P	0.047	0.699	274	27	1.000	1.000	No	No
pHwater	EC	-0.046	0.703	275	26	1.000	1.000	No	No
N:P	Mg^{2+}	0.044	0.716	276	25	1.000	1.000	No	No
C:P	Mn	0.044	0.718	277	24	1.000	1.000	No	No
Clay	Ca^{2+}	-0.043	0.722	278	23	1.000	1.000	No	No
pHKCl	CEC	-0.043	0.726	279	22	1.000	1.000	No	No
C:P	Sand	0.041	0.738	280	21	1.000	1.000	No	No
CEC	Mn	0.038	0.755	281	20	1.000	1.000	No	No
Mg^{2+}	P	0.036	0.765	282	19	1.000	1.000	No	No

C:N	Base saturation	0.036	0.767	283	18	1.000	1.000	No	No
K^+	OC	0.029	0.808	284	17	1.000	1.000	No	No
CEC	ESP	0.028	0.817	285	16	1.000	1.000	No	No
pHwater	Silt	-0.027	0.821	286	15	1.000	1.000	No	No
ESP	N	-0.027	0.821	287	14	1.000	1.000	No	No
ESP	Zn	-0.025	0.836	288	13	1.000	1.000	No	No
K^+	Fe	0.024	0.841	289	12	1.000	1.000	No	No
Mn	N	0.022	0.855	290	11	1.000	1.000	No	No
C:P	K^+	-0.019	0.875	291	10	1.000	1.000	No	No
C:N	N	-0.017	0.886	292	9	1.000	1.000	No	No
pHKCl	SAR	0.016	0.895	293	8	1.000	1.000	No	No
Mg^{2+}	Mn	0.016	0.898	294	7	1.000	1.000	No	No
Sand	P	0.013	0.912	295	6	1.000	1.000	No	No
Silt	Mn	-0.013	0.915	296	5	1.000	1.000	No	No
C:N	K^+	-0.009	0.944	297	4	1.000	1.000	No	No
Ca^{2+}	P	-0.006	0.959	298	3	1.000	1.000	No	No
SAR	N	-0.004	0.971	299	2	1.000	1.000	No	No
C:N	pHKCl	0.003	0.983	300	1	0.983	1.000	No	No